Gen AI with Element

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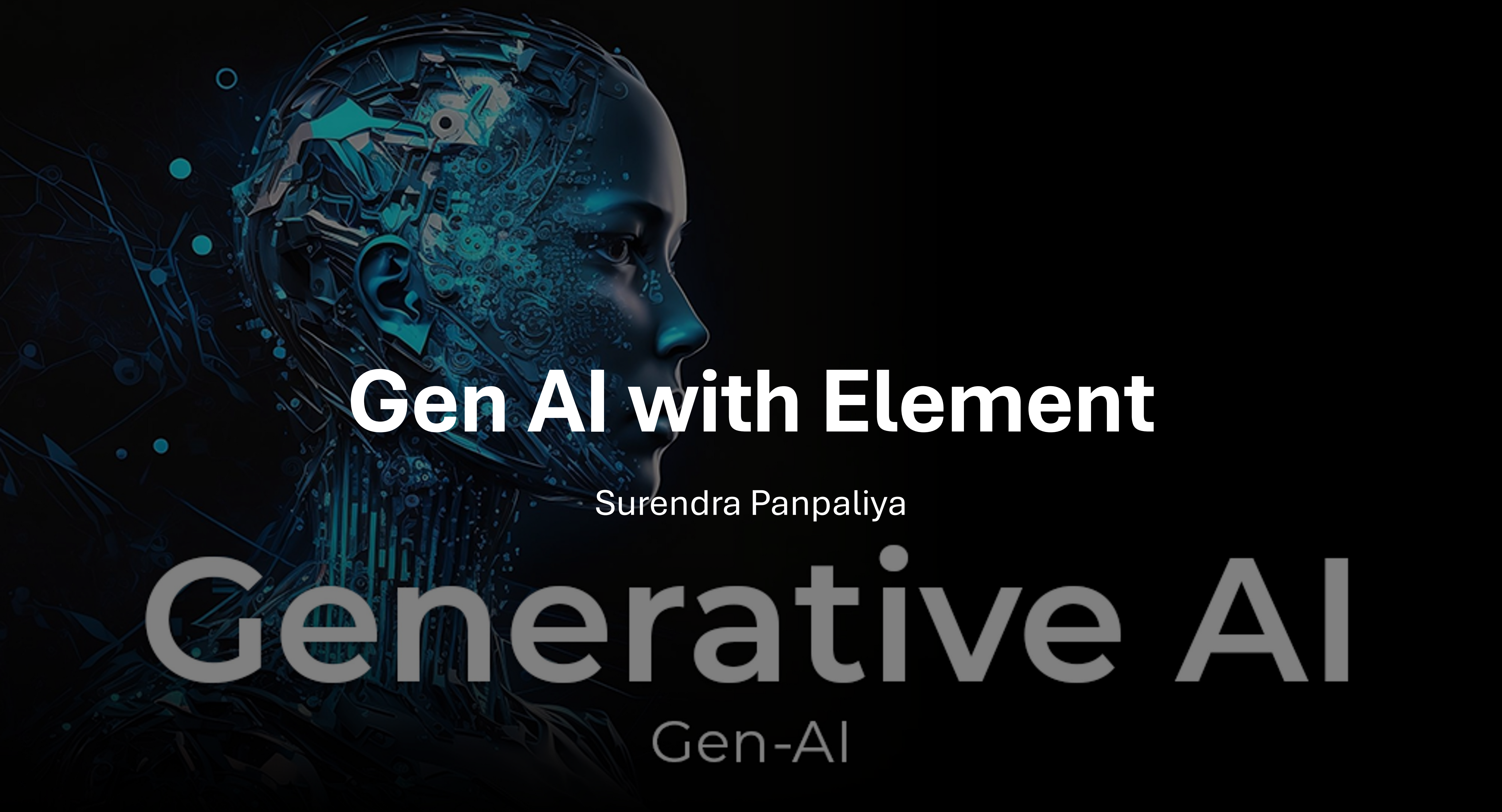


PREREQUISITES

Participants should have:

Basic knowledge of **Python programming**

Familiarity with **APIs**, **JSON**, and **HTTP requests**

General understanding of **Machine Learning** and **NLP concepts**

PREREQUISITES

- Awareness of **cloud platforms** (Azure or GCP preferred)
- Prior exposure to **Jupyter Notebooks** or **VS Code**
- Knowledge of **RESTful APIs**, **Docker**, and **Git**
- Some experience with **LLMs** or **prompt engineering**

LAB SETUP REQUIREMENTS

Python 3.10+ installed (preferably in a virtual environment)

JupyterLab or **VS Code** with Python plugin

Access to:

Azure OpenAI API key (or)

Google GenAI credentials (Vertex AI Studio, PaLM/Gemini API)

📌 LEARNING OUTCOMES

Explain key concepts in

Generative AI and

Transformer-based LLMs

Build and interact with

OpenAI/Gemini models using Python

📌 LEARNING OUTCOMES

Design **RAG pipelines** integrated with

LangChain + Milvus

Apply structured prompt

engineering strategies in LLM apps

📌 **LEARNING OUTCOMES**

Utilize Walmart's internal

LLM Gateway and **evaluation platforms**

Create simple **Agentic applications**

with planning, execution, and tools

📌 LEARNING OUTCOMES

- Troubleshoot common LLM issues
- such as hallucination or bias
- Build and present a functional
- **Excel-based report builder** GenAI app

Agenda

DAY 1: GENAI FOUNDATION & ARCHITECTURE

DAY 2: WALMART GENAI ECOSYSTEM

DAY 3: APPLICATION DEVELOPMENT WITH GENAI

DAY 4: HACKATHON & DEPLOYMENT

DAY 3: APPLICATION DEVELOPMENT WITH GENAI

Objective:

Build enterprise-level GenAI apps

using agent-based designs,

chaining, and governance features.

Agenda

- Agentic AI & Task Chaining
- Governance and Evaluation
- Hands-On Session

Objective

BUILD ENTERPRISE-LEVEL GENAI APPS

USING AGENT-BASED DESIGNS,

CHAINING, AND

GOVERNANCE FEATURES.

Agentic AI & Task Chaining

What is Agentic AI?

Core components:

Agents

Tools

Planner

Executor

What is Agentic AI?

- Agentic AI = Smart AI agents that
- Understand goals
- Plan steps
- Execute tasks
- Work together (like a team)

What is Agentic AI?

New way of using AI

to **plan, execute, and manage tasks**

independently

like an intelligent assistant

that thinks and acts.

What is Agentic AI?

ACT LIKE A RESPONSIBLE ASSISTANT,

DOING TASKS **INDEPENDENTLY**

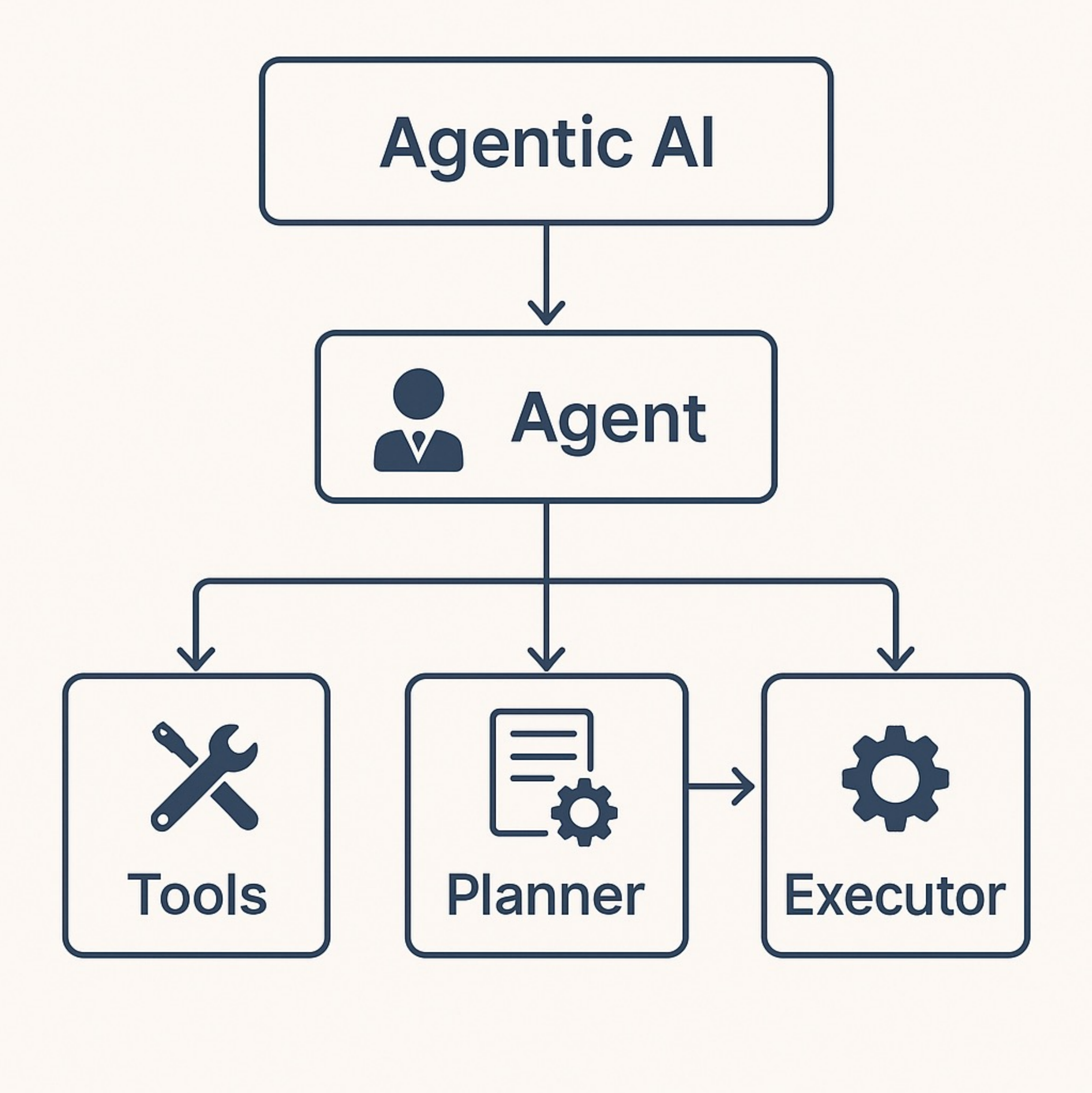
BASED ON GOALS YOU GIVE IT.

Agentic AI uses Agents that can

- Understand goals
- Break them into tasks
- Use tools or APIs to complete tasks
- Adjust actions based on results

Core Components of Agentic AI

1. Agents
2. Tools
3. Planner
4. Executor


Agentic AI
Agent
Tools
Planner
Executor

1. Agents

Individual units with specific roles or goals.

Can reason, learn, and make decisions

based on their goals.

2. Tools

Resources or capabilities available to agents.

For example: APIs, databases,

web search engines, or AI models.

3. Planner

Determines how to achieve the goal.

Decides which agents

should collaborate, and in what order.

4. Executor

Executes the planned tasks.

Coordinates interaction

between agents and tools

to complete tasks.

Core Components of Agentic AI

Component	Purpose
🧑‍💼 Agent	An intelligent entity that receives instructions and decides what to do.
🧰 Tool	A function or API the agent can use (like a search engine, database query, code interpreter, etc.)
📋 Planner	Breaks down a big problem into smaller tasks.
⚙️ Executor	Executes tasks one by one, using tools, and adjusts based on feedback.

Walmart-Specific Analogy

Imagine an **AI Assistant**

for Inventory Management at Walmart

Goal:

Restock top-selling items in Pune region.

Walmart-Specific Analogy

Role	Example Behavior
Agent	Decides what to do, e.g., "I need to find the top-selling items, then check current inventory, then create purchase orders."
Tool	Calls internal APIs like get_sales_data(), check_inventory(), create_PO()

Walmart-Specific Analogy

Role	Example Behavior
Planner	Breaks it into: 1. Identify top items 2. Check stock 3. Restock
Executor	Runs the steps, checks outputs, and moves to the next step

Real-world Example for Walmart

Imagine Walmart wants an AI solution

to automatically handle

customer inquiries about products

Real-world Example for Walmart

Agent	Role/Goal	Tools
Chat Agent	Interacts with customers	ChatGPT, Dialogflow
Product Agent	Finds product information	Walmart product catalog API
Stock Agent	Checks stock availability	Walmart inventory database
Pricing Agent	Provides current pricing details	Pricing API

Agent Design Patterns

ReAct

Chain-of-Thought

Plan-and-Execute

What Are Agent Design Patterns?

AGENT DESIGN PATTERNS DEFINE

HOW AN AI AGENT

THINKS, PLANS, AND

EXECUTES TASKS.

What Are Agent Design Patterns?

These patterns guide:

- How the agent reasons
- How tasks are broken down
- How tools are used
- How the agent adapts to feedback

Key Agent Design Patterns

- ReAct (Reasoning and Acting)
- Chain-of-Thought (CoT)
- Plan-and-Execute (PnE)

ReAct (Reasoning and Acting)

The agent **reasons**

step-by-step and then

decides an action,

repeating until

the final answer is found.

Example

“Which supplier provides the best rate for top-selling laptops?”

Thinks:

I need to fetch top-selling laptops

→ compare supplier prices → choose best.

Acts: Calls APIs or tools one step at a time.

Pattern

Thought → Action → Observation →

Thought → Action → … → Final Answer

Chain-of-Thought (CoT)

The agent is encouraged

to **explain its reasoning process**

before giving an answer.

It doesn't act with tools

but reasons clearly.

What is Chain-of-Thought (CoT)?

Instead of jumping to the final answer,

the model is instructed to **think aloud**

by breaking the problem into

logical reasoning steps.

What is Chain-of-Thought (CoT)?

Improves accuracy, interpretability

often factual correctness,

especially for complex or

multi-step questions.

Example

If Walmart has 20% more customers this month,

what could be the reason?

Example

Agent explains:

"Footfall has increased."

"Marketing campaign launched last month."

"Competitor store closed nearby."

Prompt Example

prompt = "Walmart's sales increased by 20%.

Think step-by-step about possible causes."

response = llm(prompt)

Plan-and-Execute (PnE)

The agent **first plans**

the full sequence of tasks,

then executes each step.

Example (Walmart Use Case)

“Launch a weekend sale campaign

across 3 top-selling categories.”

Example (Walmart Use Case)

Plan	Plan: Identify top categories
Pricing	Fetch pricing
Create	Create campaign draft
Submit	Submit for approval
Execute	Execute: Each step is called via API or tool.

What is Plan-and-Execute (PnE)?

Powerful agent pattern that:

First plans a sequence of subtasks

based on the user request.

Then **executes each step**,

often using tools, APIs, or functions.

What is Plan-and-Execute (PnE)?

This improves

reliability,

transparency, and

modularity in AI task solving.

Use Case

Prompt:

“Launch a weekend sale campaign

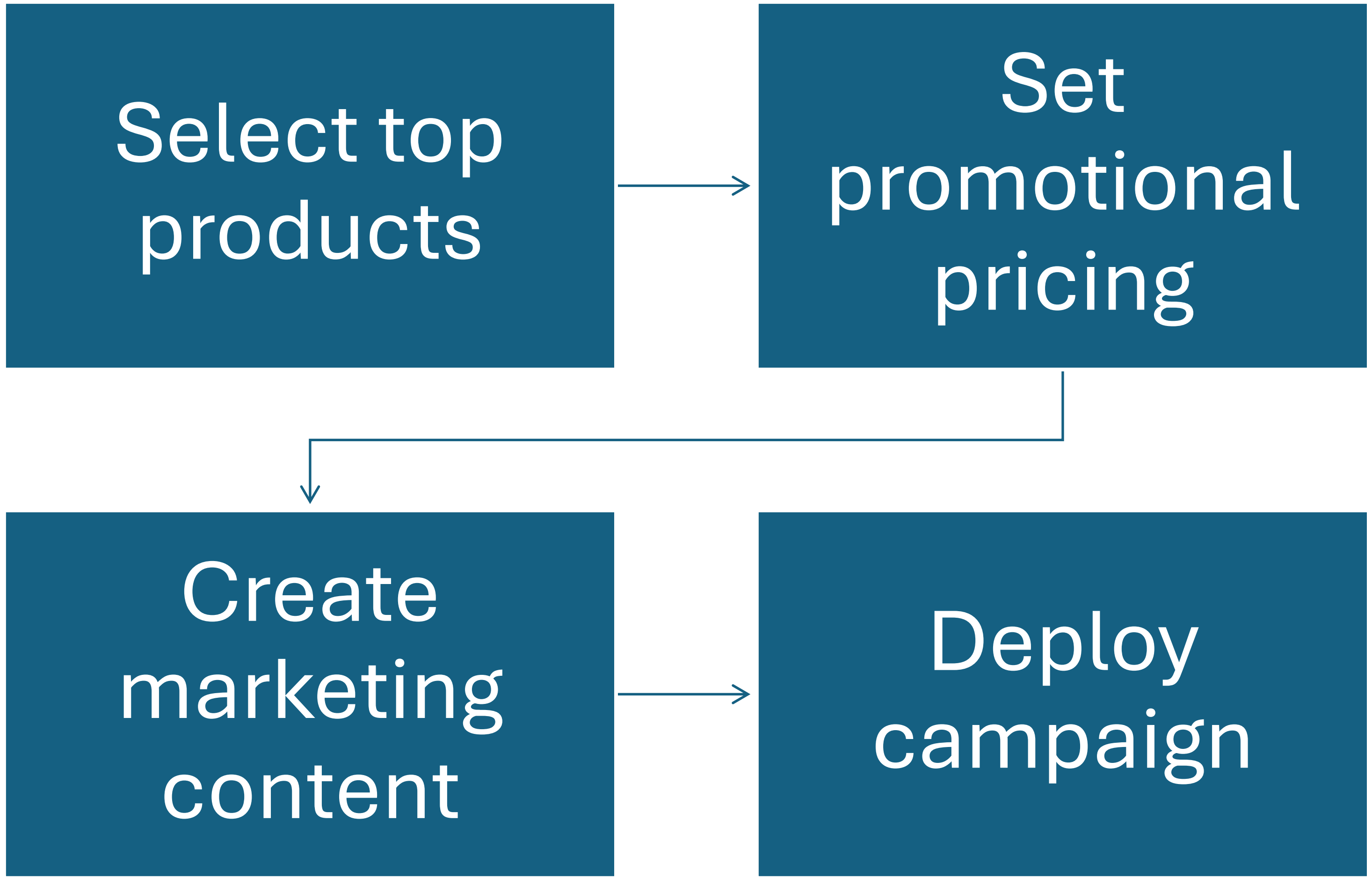
across 3 top-selling categories.”

Use Case

Launching promotional campaigns.

Restocking workflows.

Example Plan



PnE vs CoT vs ReAct

Pattern	Reasoning Style	Complexity	Ideal Use Cases (Walmart)
ReAct	Iterative reasoning-action loop	Medium	Inventory, pricing decisions
Chain-of-Thought	Step-by-step reasoning	Low	Sales trend analysis, customer insights
Plan-and-Execute	Detailed planning & structured execution	Medium-High	Promotional campaigns, restocking workflows

PnE vs CoT vs ReAct

Pattern	Suitable for	Reasoning type	Complexity
ReAct	Dynamic decision-making with tools	Reason and act iteratively	Medium
Chain-of-Thought	Complex analytical tasks	Step-by-step logical reasoning	Low
Plan-and-Execute	Structured multi-step workflows	Planning ahead, then executing	Medium-High

PnE vs CoT vs ReAct

Style	Planning	Tool Use	Ideal For
CoT	Yes	❌	Pure reasoning/explanation
ReAct	Step-by-step	✅	Interactive tool calls while thinking aloud
Plan-and-Execute	✅ Full upfront plan	✅ Sequential tool calls	Multi-step automation, real tasks

Governance and Evaluation

- Walmart GenAI Evaluator: Why evaluation is critical
- Standardization of LLM outputs
- Monitoring hallucination, bias, and failures
- Overview of Walmart's Agentic AI platform offerings

What is GenAI Governance?

- Set of practices ensuring that
- AI solutions comply with organizational policies,
- legal frameworks, ethical standards,
- and business objectives.

What is GenAI Evaluation?

Systematic assessment

to ensure AI models are accurate,

fair, reliable, robust, and

aligned with business requirements.

What is GenAI Evaluation?

- Evaluation is a structured approach
- to measure and improve the effectiveness,
- accuracy, fairness, and safety
- of Generative AI (GenAI) models.

Why Is GenAI Evaluation Essential at Walmart?

- Ensuring Accuracy & Reliability
- Mitigating Risks & Biases
- Enhancing Customer Trust
- Regulatory Compliance
- Continuous Improvement

1️⃣ Ensuring Accuracy & Reliability

Confirms the AI-generated responses and

actions align with Walmart's business requirements.

Maintains trust by providing consistently accurate results.

2️⃣ Mitigating Risks & Biases

- Detects and reduces unwanted biases
- that could harm Walmart's reputation.
- Prevents incorrect decisions
- that could lead to financial or operational risks

3️⃣ Enhancing Customer Trust

- Customers interact confidently
- with reliable, transparent AI solutions.
- Strengthens Walmart’s brand value
- by ensuring fairness and trustworthiness
- in automated interactions.

4️⃣ Regulatory Compliance

Ensures that Walmart’s AI solutions

comply with global regulations and

standards, avoiding legal issues.

5️⃣ Continuous Improvement

Provides insights into

model performance,

highlighting areas

for further development.

5️⃣ Continuous Improvement

Enables Walmart

to maintain competitive

advantage through adaptive and

improved GenAI solutions.

Key Metrics in GenAI Evaluation

Accuracy & Precision:

Correctness of AI responses.

Fairness & Bias:

AI decisions equitable across diverse user groups.

Key Metrics in GenAI Evaluation

Robustness:

Stability of AI under various conditions.

Safety & Ethics:

AI adherence to ethical guidelines and policies.

Potential Risks Without Effective Evaluation

- Misleading customer interactions.
- Financial losses from incorrect AI decisions.
- Reputational damage from
- biased or inappropriate outputs.
- Legal and compliance risks.

Practical Steps for Walmart GenAI Evaluators

- Set clear, measurable criteria aligned with business objectives.
- Implement standardized evaluation frameworks and tools.
- Conduct regular audits and reviews.
- Use feedback loops for continuous refinement

Conclusion

At Walmart,

rigorous GenAI evaluation

is not optional

it’s foundational.

Conclusion

It directly impacts

customer trust,

brand reliability,

business profitability, and compliance.

Conclusion

Prioritizing GenAI evaluation ensures

Walmart remains a leader in

ethical, efficient, and

effective AI deployment.

MCP Server using LangChain

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Model Context Protocol (MCP)

Standardized framework

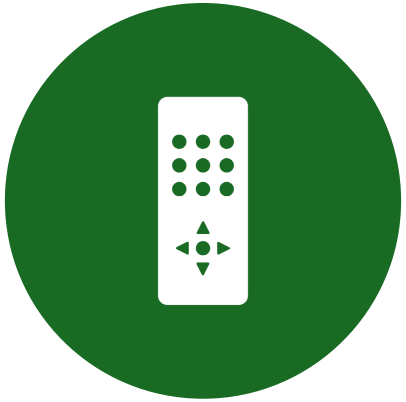
Allows **AI models**

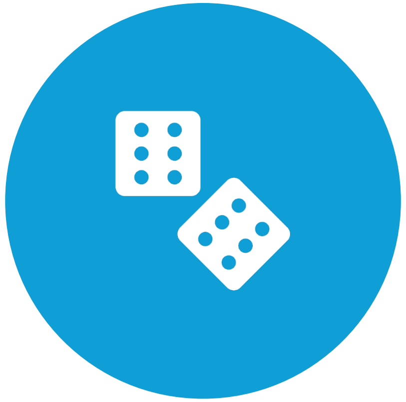
To **interact with real-world tools,**

APIs, and data sources

in a **safe, modular, and controlled way**.

What is MCP (Model Context Protocol) Server?





LIKE A **TOOLBOX AND** **REMOTE CONTROL** FOR AI MODELS.

What is MCP (Model Context Protocol) Server?

Lets AI systems not just think and answer

but also **take real-world actions**

by calling APIs, tools, or databases

in a safe, modular way.

MCP Workflow

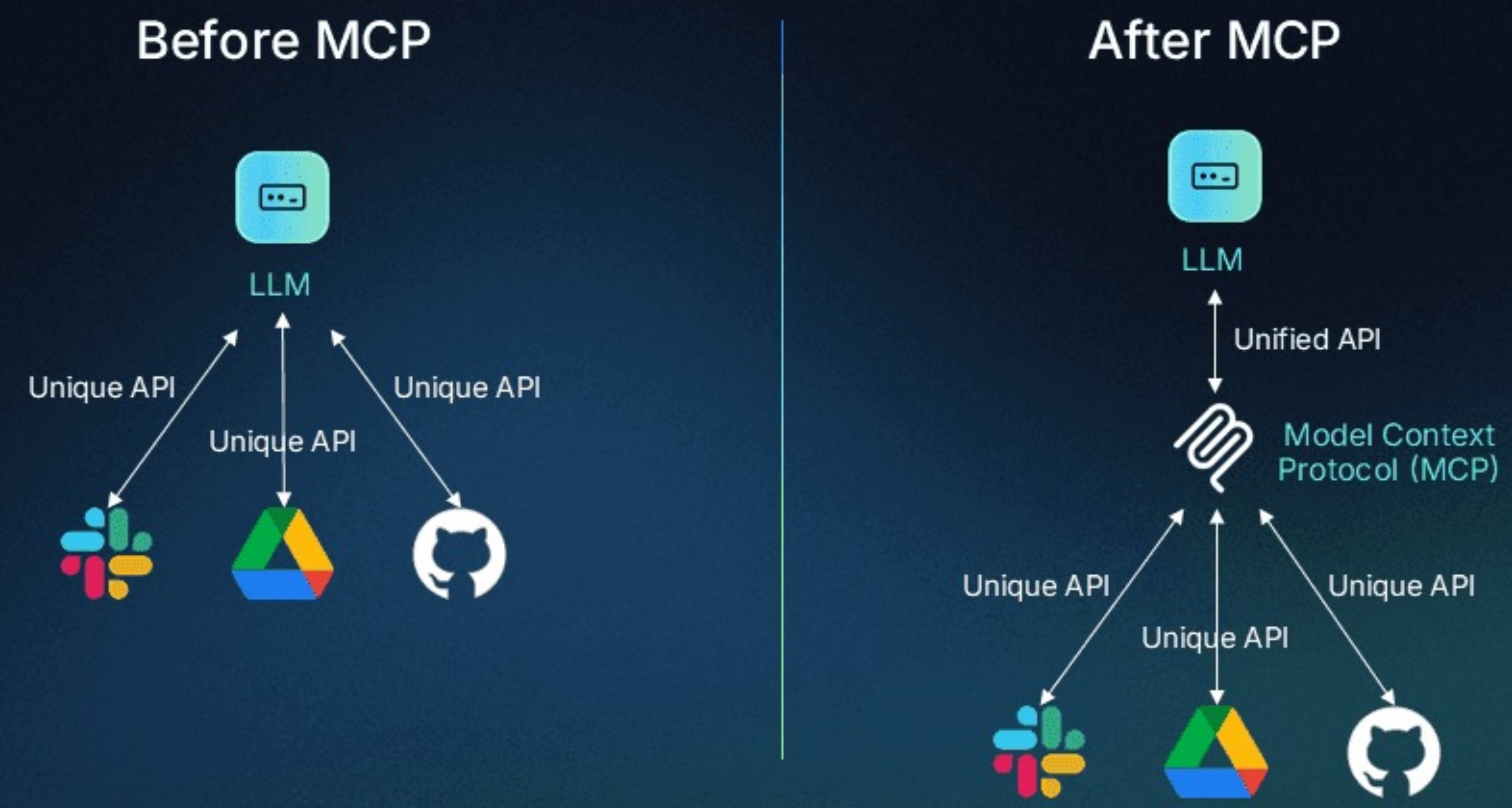
[AI Model / LLM]

↓

[MCP Protocol Layer]

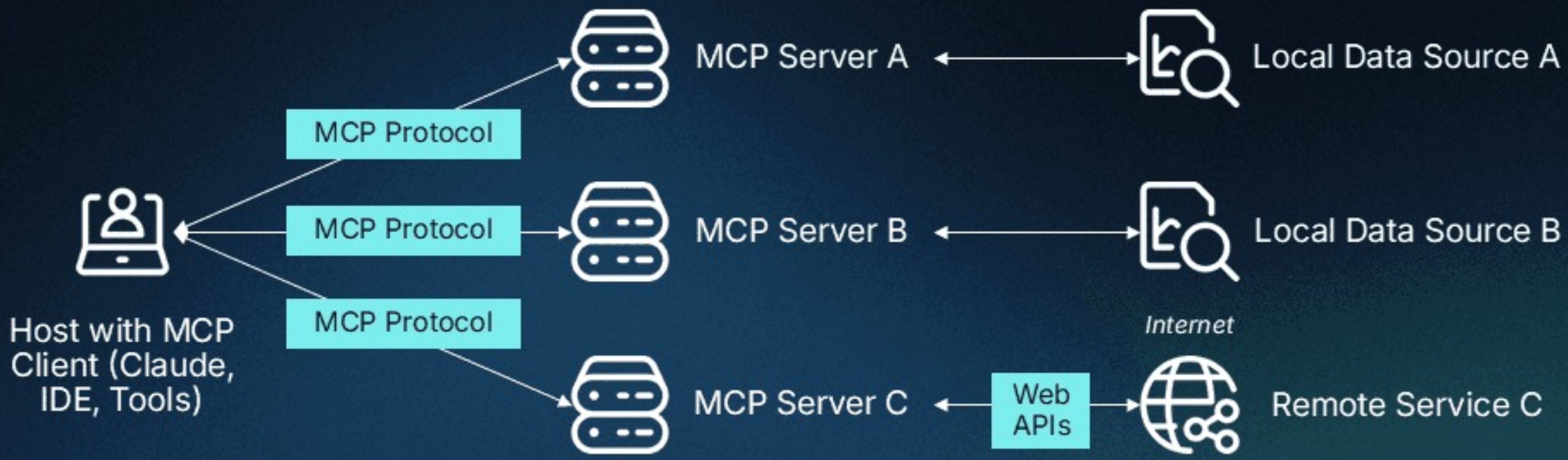
↓

[Tools / APIs / Databases / Systems]


descope
Before MCP
LLM
Unique API
Unique API
Unique API
After MCP
LLM
Unified API
Model Context
Protocol (MCP)
Unique API
Unique API
Unique API

MCP Architecture

descope



MCP Core Components

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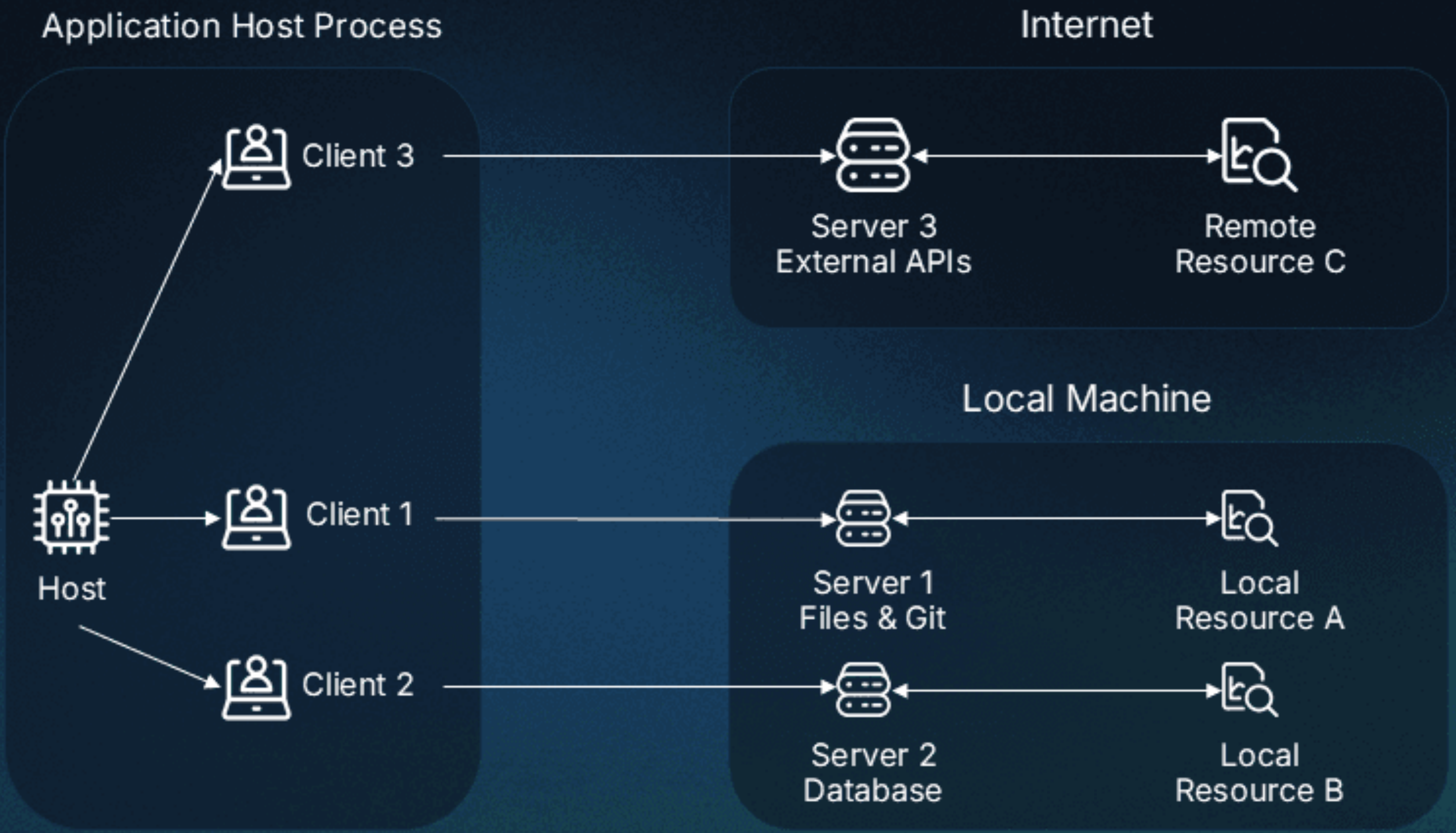

Why Do We Need MCP Server?

Large Language Models

Can **think**, **reason**, **write**, and **chat**.

But **cannot directly act**

(e.g., change a price, check stock, place an order).

Why Do We Need MCP Server?

MCP Server bridges

this gap by letting AI interact

with the real world

securely and efficiently.

Everyday Analogy

Role	Real Life	AI Example
Brain (LLM)	Decides to call a friend	Decides to change product price
MCP (Hands/Phone)	Dials the number	Calls Walmart’s pricing system to change the price

Use Cases of MCP in Walmart

Dynamic Pricing Agent

Problem	Competitors change prices frequently. Walmart wants real-time price adjustments.
With MCP	AI analyzes competitor prices, decides on new pricing, and calls Walmart's Pricing API via MCP to update the catalog.

Inventory Check & Reorder Agent

Problem	**Store managers need real-time inventory updates and automatic reorder suggestions.**
With MCP	AI checks stock using Walmart's Inventory API via MCP and triggers reorder workflows if stock is low.

Order Management Assistant

Problem	**Customer service agents spend time checking order statuses or handling cancellations.**
With MCP	AI accesses Walmart's Order Management System (OMS) via MCP to check order status, process returns, or cancel orders automatically.

Personalized Marketing Agent

Problem	**Sending personalized offers requires accessing customer history, preferences, and generating content.**
With MCP	AI pulls customer data via secure MCP calls, generates personalized offers using GenAI, and sends promotions using marketing APIs.

Store Operations Copilot

Problem	**Store managers handle multiple daily operational decisions manually.**
With MCP	AI agents gather sales data, competitor analysis, and staffing needs via MCP and provide daily actionable recommendations.

Walmart-Specific Use Case

Scenario: Dynamic Pricing Agent

Problem:

Walmart wants to adjust prices in real-time based on:

Competitor pricing

Stock levels

Demand trends

Solution with MCP

Step	Who Does It?
AI Agent decides: "Lower price to ₹445"	LLM + LangChain
Actual price update in Walmart system	MCP Server calls Pricing API

RAG vs Agentic AI vs MCP

Feature / Aspect	RAG	Agentic AI	MCP
Main focus	Retrieval + grounded generation	Planning, reasoning, tool orchestration	Standardized context & tool interface
Handles multi-step tasks	❌	✅	❌ (but can be used by agents)
Requires vector DB	Usually	Optional	Optional

RAG vs Agentic AI vs MCP

Feature / Aspect	RAG	Agentic AI	MCP
Tool/API integration	Minimal	Core feature	Yes, as standardized MCP tools
Interoperability	❌	❌ (custom per agent)	✅ cross-app/LLM
Example in Walmart	Return policy Q&A	Return eligibility + logistics	Serve policy DB & inventory API to any LLM client

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Happy Learning !! Thanks for Your Patience ☺

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